1. Problem Statement and Objectives

The institution wants to understand factors associated with student dropout and academic success, and predict the eventual outcome Dropout, Enrolled or Graduate. Objectives include data-quality checks, descriptive statistics, visualizations, hypothesis testing, multiclass classification, model interpretation and responsible recommendations.

2. Methodology

The semicolon-separated dataset contains 4,424 records and 37 columns. *Target* is the outcome variable. Integer-coded attributes such as course, application mode and qualification are categories, not continuous measurements. Missing values, duplicate rows, ranges and target labels were checked. Numerical fields are median-imputed and standardized; coded categorical fields are mode-imputed and one-hot encoded.

Leakage prevention: the primary model predicts at enrollment, so all 12 first- and second-semester curricular-unit variables were excluded from its predictors. They are examined separately in EDA and hypothesis testing. A stratified 75/25 train-test split and class-weighted logistic regression were used.

3. Statistical Findings

The target classes are Graduate: 2,209 (49.93%), Dropout: 1,421 (32.12%), and Enrolled: 794 (17.95%). Graduate is the largest class and Enrolled is the smallest, so accuracy alone is insufficient.

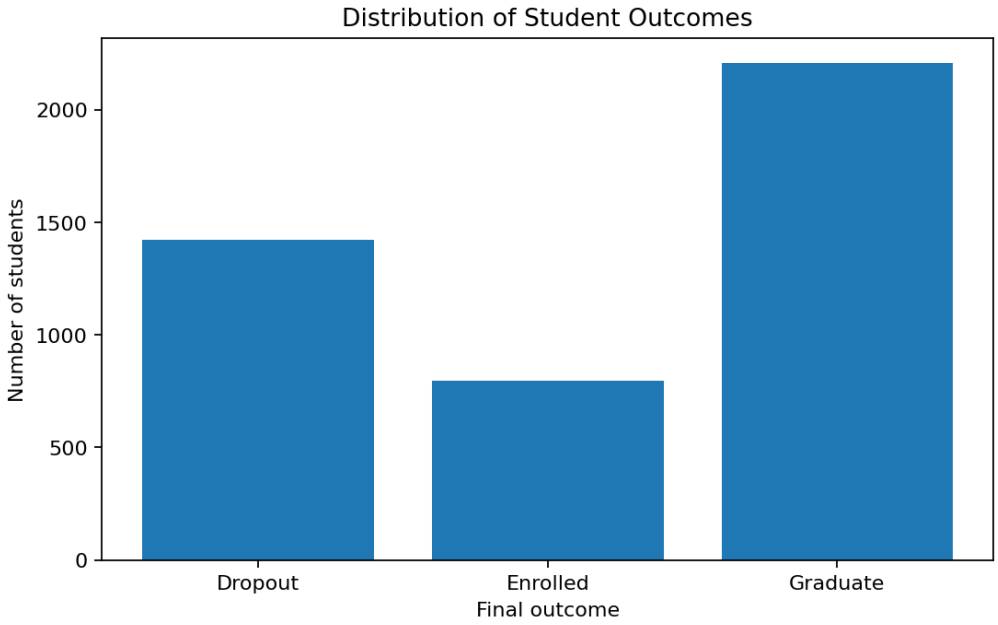


Figure 1. Counts of Dropout, Enrolled and Graduate outcomes. The class distribution is imbalanced.

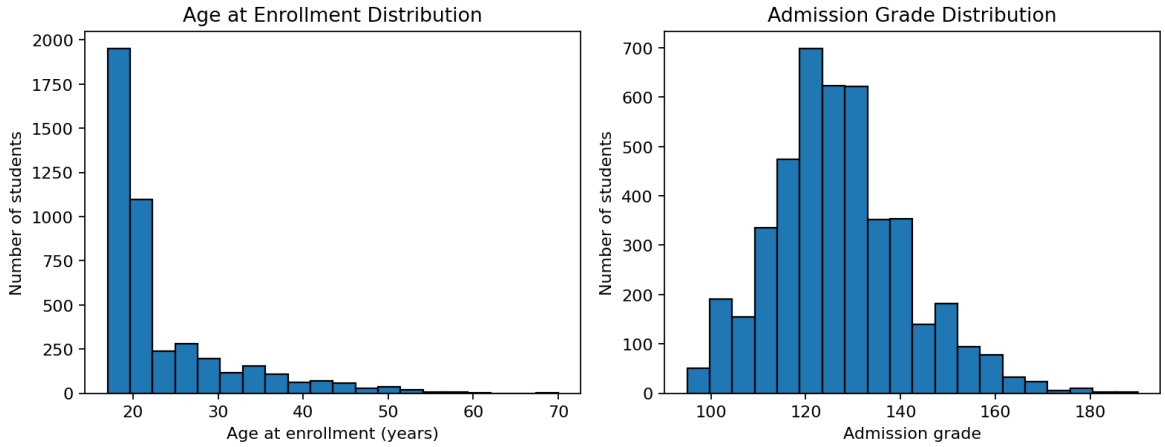


Figure 2. Histograms of age at enrollment and admission grade. Age ranges from 17 to 70; distributions show variation and a right tail in age.

Mean admission grades by outcome are Dropout 124.96, Enrolled 125.53, and Graduate 128.79. The Graduate group has the highest mean, though the groups overlap.

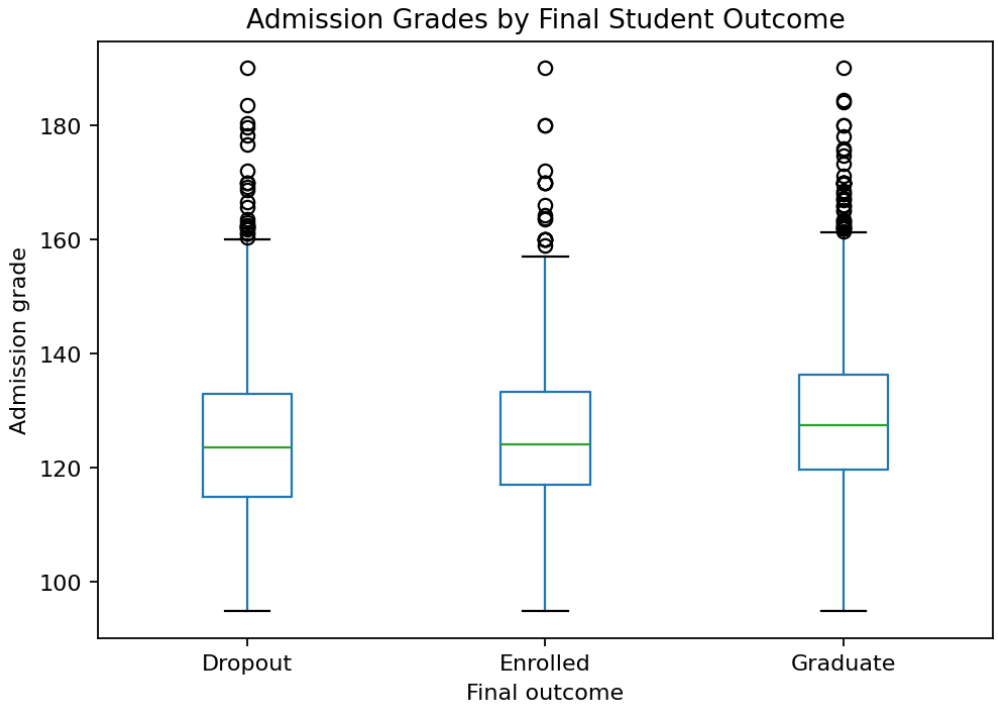


Figure 3. Admission grades by final outcome. Compare medians, spread and overlap; this association does not establish causality.

Mean first-semester grades are Dropout 7.26, Enrolled 11.13, and Graduate 12.64. The dropout group has a notably lower average.

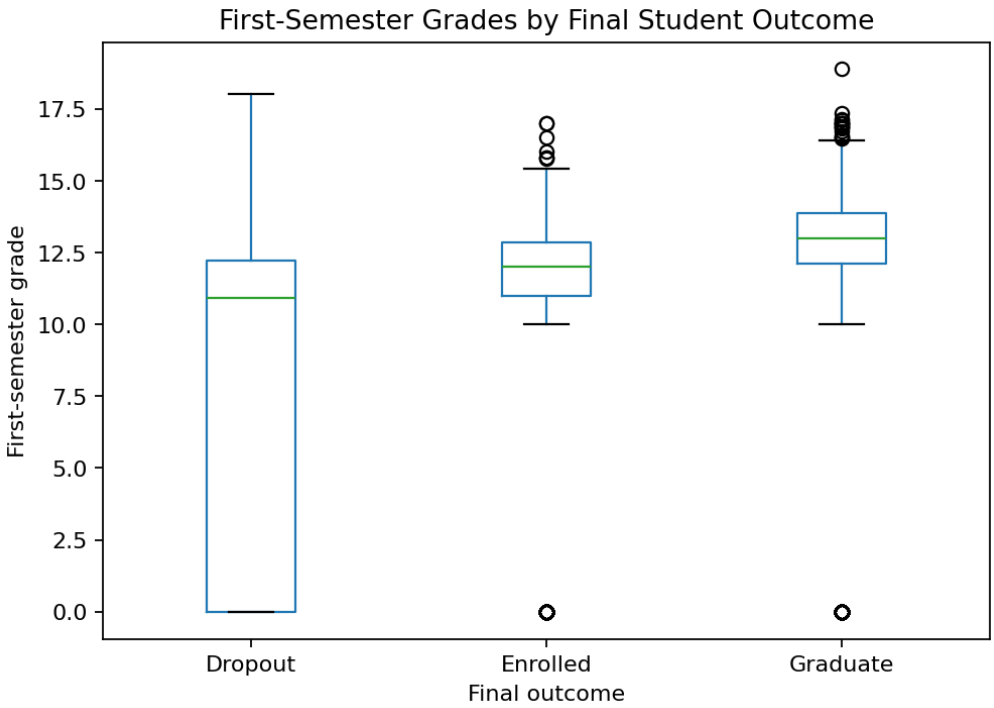


Figure 4. First-semester grade distributions by eventual outcome. These grades may support semester-one monitoring but are excluded from the enrollment-time model.

4. Age and Outcome Composition

Outcome composition differs across age groups. The dropout share is higher in older age bands than among students aged 20 or younger. This suggests a topic for further investigation, not a reason to assume age causes dropout.

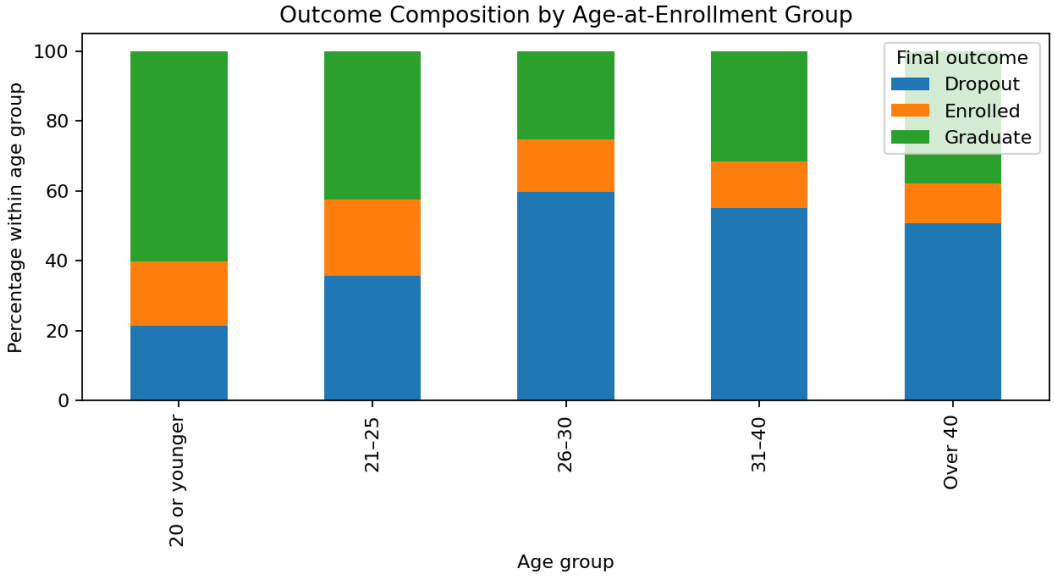


Figure 5. Stacked bars show the percentage of outcomes within each age group; each bar sums to 100%.

5. Hypothesis Test

H_0: first-semester grade distributions are the same across Dropout, Enrolled and Graduate. H_1: at least one distribution differs. The Kruskal–Wallis test compares three independent groups without requiring normality; observations should be independent, and very different distribution shapes limit median-only interpretations.

At α = 0.05, Kruskal–Wallis H = 1094.186 and p = 2.51e-238. Reject H_0: there is strong statistical evidence of differences in first-semester grade distributions. Approximate epsilon-squared = 0.247, indicating a non-trivial group association in this sample. Statistical significance does not prove causation.

6. Model Evaluation

On the held-out test set of 1,106 students, accuracy = 0.575; macro precision = 0.561, macro recall = 0.560, and macro F1 = 0.549. Macro metrics give each class equal weight.

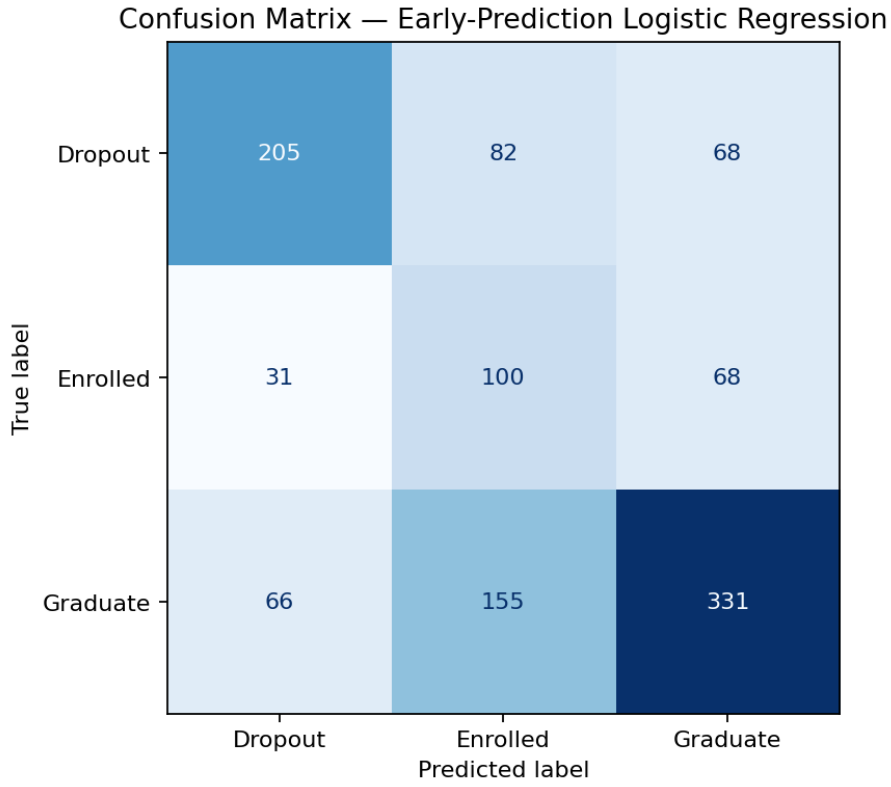


Figure 6. Confusion matrix. Rows are actual outcomes and columns are predicted outcomes; off-diagonal cells represent classification errors.

Class	Precision	Recall	F1-score	Support
Dropout	0.679	0.577	0.624	355
Enrolled	0.297	0.503	0.373	199
Graduate	0.709	0.600	0.650	552
Macro avg	0.561	0.560	0.549	1106

F1 is higher for Graduate (0.650) and Dropout (0.624) than for Enrolled (0.373). Enrolled precision is only 0.297, so many predictions of Enrolled are incorrect. Class weighting improves attention to

smaller classes but does not eliminate errors.

7. Model-Derived Feature Interpretation

Dropout: largest positive class coefficients include Mother's occupation_0 (1.50), Father's occupation_90 (1.05), Mother's occupation_90 (1.04), Mother's qualification_9 (0.86). Coefficients represent conditional associations in the fitted model; coded-category coefficients depend on reference coding and are not causal effects.

Enrolled: largest positive class coefficients include Father's qualification_39 (1.11), Course_9119 (0.89), Mother's occupation_192 (0.88), Application mode_5 (0.83). Coefficients represent conditional associations in the fitted model; coded-category coefficients depend on reference coding and are not causal effects.

Graduate: largest positive class coefficients include Application mode_15 (1.35), Marital status_5 (0.87), Nacionality_11 (0.78), Father's qualification_43 (0.64). Coefficients represent conditional associations in the fitted model; coded-category coefficients depend on reference coding and are not causal effects.

8. Limitations

- The model predicts recorded historical outcomes and may not generalize to another institution or future cohort.
- The early model excludes semester performance to avoid leakage, so it may perform less well than a model built after semester one.
- Class imbalance and low Enrolled precision mean class-specific errors require attention.
- Logistic regression may miss nonlinear relationships and complex interactions.
- Demographic associations can reflect structural inequities and must not be used to label or penalize individuals.
- The target is status at the end of normal course duration, not necessarily eventual long-term completion.

9. Recommendations

- Offer voluntary tutoring, advising, academic planning and financial-aid guidance to students who may benefit.
- Use first-semester performance for a separately validated early-warning process and timely supportive outreach.
- Investigate scheduling, work, family, financial and resource barriers rather than attributing dropout to demographic traits.
- Audit errors and performance across relevant student groups, provide human review and allow records to be corrected.
- Evaluate support interventions prospectively to learn whether they improve outcomes.

10. Conclusion

The dataset has 4,424 students and 37 columns. Graduate is the largest class (49.93%) and Enrolled the smallest (17.95%). First-semester grade distributions differ significantly ($H = 1094.186$, $p < 0.001$). The early-prediction logistic regression achieved 0.575 accuracy and 0.549 macro F1, with weaker performance on Enrolled. The model may guide supportive outreach and further investigation, but it is not causal evidence and should not be used for punitive decisions.

Submission note: keep Studentdata.csv beside the notebook, run all cells, fill in the title page and verify that you understand the results and methods. Follow your faculty's attribution rules.